

Covariance-aware Bayesian constraints on power-law Palatini $f(\mathcal{R})$ cosmology

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Abstract

We constrain the dust-filled Palatini model $f(\mathcal{R}) = \mathcal{R} - \beta\mathcal{R}^{-n}$ with 32 cosmic-chronometer measurements, including their available systematic covariance, and the Pantheon+ Hubble diagram with its released full statistical-plus-systematic covariance, together with the 13-element DESI DR2 BAO likelihood. Present Friedmann normalization derives $\tilde{\beta} = \beta/H_0^{2(n+1)}$ from each (n, k) , where $k = \kappa\rho_{m0}/(3H_0^2) = \Omega_{m0}$. We perform static DYNESTY 3.1.0 nested sampling with 1024 live points and $\Delta \ln Z = 0.01$. The nonlinear (n, k) posterior is sampled directly, while the conditionally Gaussian H_0 , supernova magnitude offset, and uncalibrated BAO scale $A_{\text{BAO}} = c/(H_0 r_d)$ are analytically marginalized over explicit bounded priors and reconstructed by conditional posterior draws. Thus DESI constrains the late-time distance–redshift shape without importing a Λ CDM sound horizon or claiming a self-consistent early-Universe Palatini calculation. The SH0ES-anchored fit is our direct late-Universe Palatini analysis. We also condition a separate run on the published Planck base- Λ CDM H_0 summary solely as a comparative baseline. Because that H_0 value is inferred from CMB data assuming base Λ CDM, and we do not evaluate the Planck CMB likelihood in Palatini gravity, this comparison is not a self-consistent Planck constraint on the Palatini model or its parameters. The SH0ES fit and the Planck-conditioned baseline give $\tilde{\beta} = 3.495^{+0.450}_{-0.387}$, $n = -0.098^{+0.067}_{-0.067}$, $k = 0.287^{+0.014}_{-0.013}$ and $\tilde{\beta} = 3.373^{+0.431}_{-0.380}$, $n = -0.116^{+0.067}_{-0.067}$, $k = 0.287^{+0.014}_{-0.015}$, respectively, for uniform priors $-1 \leq n \leq 1$ and $0 \leq k \leq 1$; intervals are posterior 16th–84th percentiles around the median. Relative to otherwise matched analyses without DESI, the BAO likelihood reduces the central widths of n by factors of 4.6 and 5.0 and those of k by factors of 7.2 and 7.0. Both central intervals lie below the exact $n = 0$ Λ CDM background limit, but the one-sided probabilities, $P(n < 0) = 0.930$ and 0.958 , constitute only modest background-level preferences. A separate DESI-only fit of (n, k, A_{BAO}) gives $\chi^2/\nu = 9.855/10$ and probability to exceed 0.453, showing an acceptable BAO shape fit. The published Gaussian anchor summaries give the standard difference-in-means statistic $Q_{\text{DM}} = 23.888$ for one degree of freedom, index of inconsistency IOI = 11.944, and $p = 1.02 \times 10^{-6}$ (4.888σ). These numbers quantify the tension already present between the two published input anchors and are not generated by the Palatini model. The posterior correlations $\rho_{nk} = 0.841$ and 0.847 show that DESI greatly contracts, but does not remove, the positive-slope degeneracy.

Keywords: Astrostatistics, Bayesian inference, scientific software, modified gravity, cosmology

1. Introduction

Replacing the Hilbert–Einstein Lagrangian by a nonlinear function of the Ricci scalar is one of the simplest extensions of general relativity. The review by Capozziello and De Laurentis summarizes metric and Palatini $f(\mathcal{R})$ theories, their scalar-tensor representations, cosmology, perturbations, and weak-field limits [1]. In the Palatini variational principle the metric and torsion-free connection are independent. The field equations therefore differ from metric $f(\mathcal{R})$ gravity, although both reduce to general relativity for a linear Lagrangian [2, 3].

We study the power-law deformation

$$f(\mathcal{R}) = \mathcal{R} - \frac{\beta}{\mathcal{R}^n}, \quad (1)$$

which has previously been confronted with supernova, expansion-rate, BAO, and CMB observations in the Palatini formalism [4, 5, 6]. An important structural point is that the present Friedmann equation removes one apparent degree of freedom. Varying β , n , and the present matter density independently without imposing this normalization overparameterizes the homogeneous background.

The present analysis upgrades the observational and statistical treatment. We use Pantheon+ with its released full covariance and a 32-point cosmic-chronometer compilation with correlated stellar-population systematics, together with the 13-observable DESI DR2 BAO likelihood [7, 8, 9, 10, 11, 12]. We perform bounded Bayesian inference with DYNESTY nested sampling [13]. The SH0ES summary provides the direct late-Universe calibration for our Palatini inference. We repeat the low-redshift fit with the published Planck base- Λ CDM H_0 summary only to construct a comparative baseline and to expose sensitivity to the adopted expansion-scale calibration. This distinction is essential: the Planck H_0 value is inferred from

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CMB data under the base- Λ CDM background and perturbation model, whereas the present work neither computes Palatini CMB anisotropies nor evaluates the Planck likelihood in Palatini gravity. The Planck-conditioned run is therefore a conditional “what-if” comparison, not a self-consistent physical Planck constraint on this Palatini theory or on $(\bar{\beta}, n, k)$. It must not be interpreted as evidence that Planck data favor or exclude any region of the reported Palatini posterior. The disagreement between the published SH0ES and base- Λ CDM Planck H_0 summaries is assessed with the index of inconsistency (IOI) and difference-in-means statistic Q_{DM} after separate fits [14, 15].

The remainder of the article is organized as follows. Section 2 develops the normalized Palatini background. Section 3 defines the Pantheon+, chronometer, DESI BAO, and alternative H_0 likelihoods. Section 4 explains analytical nuisance marginalization and nested sampling. Section 5 reports posterior constraints, the (n, k) elongation, and the IOI/ Q_{DM} comparison. Section 6 gives numerical checks and limitations, and Section 7 summarizes the conclusions.

2. Palatini background dynamics

For minimally coupled matter, the action is

$$S = \frac{1}{2\kappa} \int d^4x \sqrt{-g} f(\mathcal{R}) + S_m[g, \psi], \quad \kappa = 8\pi G, \quad (2)$$

where $\mathcal{R} = g^{\mu\nu} \mathcal{R}_{\mu\nu}(\hat{\Gamma})$ is constructed from an independent connection $\hat{\Gamma}$. Independent variations give

$$f_{\mathcal{R}} \mathcal{R}_{\mu\nu} - \frac{1}{2} f g_{\mu\nu} = \kappa T_{\mu\nu}, \quad (3)$$

$$\hat{\nabla}_\lambda (\sqrt{-g} f_{\mathcal{R}} g^{\mu\nu}) = 0. \quad (4)$$

Thus $\hat{\Gamma}$ is the Levi-Civita connection of $h_{\mu\nu} = f_{\mathcal{R}} g_{\mu\nu}$. For pressureless matter the trace is algebraic,

$$f_{\mathcal{R}} \mathcal{R} - 2f = -\kappa \rho_m. \quad (5)$$

We assume a spatially flat FLRW metric and conserved dust, $\rho_m = \rho_{m0}(1+z)^3$, and define

$$r \equiv \frac{\mathcal{R}}{H_0^2}, \quad \bar{\beta} \equiv \frac{\beta}{H_0^{2(n+1)}}, \quad k \equiv \frac{\kappa \rho_{m0}}{3H_0^2} = \Omega_{m0}. \quad (6)$$

The symbol k is the dimensionless matter coupling identifiable from the background. It is neither the FLRW spatial-curvature index nor a separate measurement of the bare constant κ .

With $\bar{f}(r) = r - \bar{\beta} r^{-n}$, the trace becomes

$$r - (n+2)\bar{\beta} r^{-n} = 3k(1+z)^3. \quad (7)$$

The existence and uniqueness of the trace root can be established directly on the sampled branch. Define $S = 3k(1+z)^3 > 0$, $A = (n+2)\bar{\beta} > 0$, and $g(r) = r - Ar^{-n} - S$. For $n \geq 0$, g is strictly increasing from a negative value to infinity. For $-1 < n < 0$, g first decreases to one minimum, which is below

$g(0^+) = -S$, and then increases to infinity; it therefore also has exactly one positive zero. At that zero, $r = S + Ar^{-n} > S$. With $y = S/r \in (0, 1)$,

$$g'(r) = 1 + n(1-y) > 0, \quad rF_r - F = -g'(r) < 0, \quad (8)$$

where the first inequality follows from $g' > 1+n$ for $-1 < n < 0$ and $g' \geq 1$ for $n \geq 0$. Moreover, $F = 1 + n(1-y)/(n+2) > 0$. The same substitution shows that $\xi > 0$: it is at least unity for $n \geq 0$, while for $n = -a$ with $0 < a < 1$,

$$\xi = 1 - \frac{3a(1-a)y(1-y)}{2[2(1-a) + ay](1-a+ay)}. \quad (9)$$

The positive denominator minus the subtractive numerator in Eq. (9) is

$$2(1-a)^2 + \frac{3}{2}a(1-a)y + \left[a^2 + \frac{3}{2}a(1-a) \right] y^2 > 0. \quad (10)$$

The Friedmann numerator is also positive. Thus the sampled open domain $-1 < n < 1$, $0 < k < 1$, and $\bar{\beta} > 0$ admits one and only one positive trace root and cannot jump to a second algebraic branch at any $z \geq 0$.

The formal $n \rightarrow -2$ case is not a regular endpoint of this branch. For finite $\bar{\beta}$, Eq. (7) reduces at $n = -2$ to the unique relation $r = S$, because the quadratic-curvature contribution cancels from the trace. However, Eq. (14) below becomes $0/0$. Normalizing the $n = -2$ Friedmann equation directly, with $x_0 = \bar{\beta} r_0 = 3k\bar{\beta}$, instead gives

$$2(k-1)x_0^2 - (5k+4)x_0 + 2(k-1) = 0. \quad (11)$$

Its discriminant is $9k(k+8) > 0$, and its two roots are negative and reciprocal. One has $-1 < x_0 < 0$ and reaches $x(z) = -1$, hence $\xi = 0$, at $z = (-1/x_0)^{1/3} - 1$; the other has $x_0 < -1$ and remains on a disconnected $\xi < 0$ branch. Thus both have $\bar{\beta} < 0$, outside the branch used here. Nor is there a regular positive- $\bar{\beta}$ continuation as $n = -2 + \epsilon \rightarrow -2^+$. Positivity of $F = 1 + (-2 + \epsilon)(1-y)/\epsilon$ forces $1-y = O(\epsilon)$; a finite limit then has $x_0 = 3k\bar{\beta} > 0$, whereas Eq. (11) permits only $x_0 < 0$. Any other scaling loses finite positive F . In the neighboring interval $-2 < n < -1$, writing $m = -n \in (1, 2)$ gives $g = r - Ar^m - S$, which has one maximum and hence can have zero, one double, or two positive roots. An analysis extending there must therefore sample $\bar{\beta}$ and a discrete branch label, test global regularity, and treat $n = -2$ as a separate model limit; simply extending the present two-parameter prior would be incorrect.

Writing $F = d\bar{f}/dr = 1 + n\bar{\beta}r^{-n-1}$ and $F_r = -n(n+1)\bar{\beta}r^{-n-2}$, the modified Friedmann equation is [5, 6]

$$E^2(z) \equiv \frac{H^2(z)}{H_0^2} = \frac{3k(1+z)^3 + \bar{f}(r)}{6F\xi^2}, \quad (12)$$

$$\xi = 1 + \frac{9}{2} \frac{F_r}{F} \frac{k(1+z)^3}{rF_r - F}. \quad (13)$$

For every trial (n, k) , a scalar root at $z = 0$ imposes $E(0) = 1$ and

$$\bar{\beta} = \frac{(r_0 - 3k)r_0^n}{n+2}. \quad (14)$$

Consequently $\bar{\beta}$ is a derived posterior quantity. The $n = 0$ limit provides an exact analytical consistency condition:

$$\bar{\beta} = 6(1 - k), \quad E^2(z) = k(1 + z)^3 + 1 - k, \quad (15)$$

which is the flat Λ CDM background.

The observables are

$$H(z) = H_0 E(z), \quad (16)$$

$$D_L(z_{\text{HD}}, z_{\text{hel}}) = \frac{c(1 + z_{\text{hel}})}{H_0} \int_0^{z_{\text{HD}}} \frac{dz'}{E(z')}, \quad (17)$$

$$\mu = 5 \log_{10} \left(\frac{D_L}{\text{Mpc}} \right) + 25. \quad (18)$$

The distinct Hubble-diagram and heliocentric redshifts in Eq. (17) follow the Pantheon+ convention.

3. Observational data and likelihood

3.1. Cosmic chronometers

The direct-expansion likelihood contains 32 cosmic-chronometer measurements over $0.07 \leq z \leq 1.965$. It includes exactly one LEGA-C-derived datum,

$$H(0.75) = 98.8 \pm 33.6 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (19)$$

from Lick-index ages of 140 massive passive galaxies in LEGA-C DR2 [16, 17, 10]. Seventeen measurements enter as independent diagonal terms. For the 15 measurements covered by the public Moresco covariance prescription, we construct

$$C_{\text{CC}} = \text{diag}(\sigma_{\text{diag}}^2) + \mathbf{u}_{\text{IMF}} \mathbf{u}_{\text{IMF}}^T + \mathbf{u}_{\text{SPS}} \mathbf{u}_{\text{SPS}}^T, \quad (20)$$

where $u_{X,i} = H_i f_X(z_i)$ and the fractional systematic curves f_X are interpolated from the released tables [9]. Equation (20) preserves correlated stellar-population-model uncertainty instead of treating every chronometer error as independent. We do not add the full-spectrum $H(0.8)$ result, whose 350-object LEGA-C DR2 parent sample overlaps the Lick-index analysis and whose authors explicitly compare the two methods on the same galaxies [18]. We also exclude the photometric $H(0.75)$ proof of concept: although its target catalog is COSMOS2015, its neural network was trained on the same 140 LEGA-C spectra and ages used by the included Borghi datum [19]. Without the cross-analysis covariances, adding either value would double count galaxy information. None of the other 31 included chronometer rows uses LEGA-C spectra.

3.2. Pantheon+

The collaboration release contains 1701 light curves of 1550 distinct Type Ia supernovae and a 1701×1701 statistical-plus-systematic covariance [7, 8]. We apply $z_{\text{HD}} > 0.01$, leaving 1590 light curves, and apply precisely the same mask to both covariance axes. The data vector is the corrected apparent magnitude $m_{B,i}^{\text{corr}}$; diagonal errors distributed for visualization are not used as an independent likelihood. The full term is

$$-2 \ln \mathcal{L}_{\text{SN}} = \Delta_m^T C_{\text{SN}}^{-1} \Delta_m + \ln |C_{\text{SN}}| + N_{\text{SN}} \ln(2\pi). \quad (21)$$

3.3. DESI DR2 baryon acoustic oscillations

We use the public DESI DR2 Gaussian combination of galaxy, quasar, and Ly α BAO measurements [11, 12]. Its 13 observables span $0.295 \leq z \leq 2.33$: one isotropic D_V/r_d measurement and six anisotropic pairs of D_M/r_d and D_H/r_d , with the released 13×13 covariance C_{BAO} . For the spatially flat background,

$$\chi(z) = \int_0^z \frac{dz'}{E(z')}, \quad D_M = \frac{c}{H_0} \chi, \quad (22)$$

$$D_H = \frac{c}{H_0 E(z)}, \quad D_V = [z D_M^2 D_H]^{1/3}. \quad (23)$$

The dust-only model does not predict the drag-epoch sound horizon r_d . We therefore do not fix r_d using Λ CDM, BBN, or CMB information. Instead every released ratio is written as

$$\mathbf{d}_{\text{BAO}}^{\text{th}} = A_{\text{BAO}} \mathbf{g}(n, k), \quad A_{\text{BAO}} \equiv \frac{c}{H_0 r_d}, \quad (24)$$

where $\mathbf{g}$ contains χ , $1/E$, or $[z\chi^2/E]^{1/3}$ in the released row order. This construction uses DESI as an uncalibrated late-time standard ruler. A derived r_d can be reported, but it is conditional on the nuisance-scale prior and is not an early-Universe prediction of the Palatini theory.

3.4. Alternative H_0 anchors

We run the identical CC, Pantheon+, and DESI likelihood separately with SH0ES, $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [20], and with the Planck 2018 base- Λ CDM summary, $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [21]. The latter is a model-dependent sensitivity anchor, not a full CMB likelihood evaluated in Palatini gravity. In particular, its Ω_m , sound-horizon, and parameter-covariance information is not imported. Because the two summaries encode discrepant calibrations, they define alternative analyses and are not multiplied into a single Gaussian likelihood.

4. Bayesian inference

4.1. Analytical nuisance marginalization

At fixed (n, k) the chronometer prediction is linear in H_0 . For one anchor $(H_{0,a}, \sigma_a)$, define

$$A_H = \mathbf{E}^T C_{\text{CC}}^{-1} \mathbf{E} + \sigma_a^{-2}, \quad (25)$$

$$B_H = \mathbf{E}^T C_{\text{CC}}^{-1} \mathbf{H} + H_{0,a} \sigma_a^{-2}. \quad (26)$$

The conditional distribution is a normal distribution with mean B_H/A_H and variance A_H^{-1} , truncated to the explicit prior $40 < H_0 < 100 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Supernovae constrain an intercept rather than H_0 and M_B separately. Writing

$$q \equiv M_B - 5 \log_{10} H_0, \quad (27)$$

and evaluating the distance shape at numerical $H_0 = 1$, the model is $\mathbf{m}^{\text{th}} = \mathbf{s}(n, k) + q \mathbf{1}$. If $\mathbf{y} = \mathbf{m}^{\text{corr}} - \mathbf{s}$, then

$$A_q = \mathbf{1}^T C_{\text{SN}}^{-1} \mathbf{1}, \quad B_q = \mathbf{1}^T C_{\text{SN}}^{-1} \mathbf{y}. \quad (28)$$

The conditional distribution of q has mean B_q/A_q and variance A_q^{-1} , truncated to $-32 < q < -25$. Cholesky solves are used throughout; the covariance matrices are never inverted explicitly.

The BAO prediction in Eq. (24) is likewise linear in its common scale. Defining

$$A_D = \mathbf{g}^\top C_{\text{BAO}}^{-1} \mathbf{g}, \quad B_D = \mathbf{g}^\top C_{\text{BAO}}^{-1} \mathbf{d}_{\text{BAO}}, \quad (29)$$

gives a conditional normal distribution for A_{BAO} with mean B_D/A_D and variance A_D^{-1} . We integrate it over the deliberately broad uniform prior $10 < A_{\text{BAO}} < 50$. Conditional posterior draws give $r_d = c/(H_0 A_{\text{BAO}})$ after pairing with the independently reconstructed H_0 draw.

At every (n, k) , all three Gaussian directions are integrated analytically, including their normalizations and truncated-normal probability masses. Conditional draws after nested sampling reconstruct H_0 , q , $M_B = q + 5 \log_{10} H_0$, A_{BAO} , and r_d . This is marginalization rather than profile optimization and therefore propagates nuisance uncertainty into the reported posterior. The nonlinear priors are uniform over

$$-1 \leq n \leq 1, \quad 0 \leq k \leq 1. \quad (30)$$

At the endpoint $n = -1$, Eq. (1) becomes the linear theory $f = (1 - \beta)\mathcal{R}$ and the present-normalization parameterization is degenerate; $k = 0$ and $k = 1$ are likewise limiting cases of the positive-matter, positive- $\bar{\beta}$ branch. These isolated endpoints have zero measure under continuous uniform priors. The sampler therefore targets the displayed priors while numerical model evaluations are restricted to their open interior.

4.2. How nested sampling works

Nested sampling rewrites the evidence $Z = \int \mathcal{L}(\theta) \pi(\theta) d\theta$ as a one-dimensional integral over remaining prior mass X . A set of live points is initially drawn from the prior. At each iteration the point with the smallest likelihood is removed, assigned a quadrature weight associated with the contracted prior mass, and replaced by a new point drawn from the prior subject to a higher likelihood threshold. Repeating this process moves the live set through nested likelihood contours. The accumulated weighted dead and live points estimate both Z and posterior expectations.

We use static DYNesty 3.1.0 with 1024 live points. The bound is multi-ellipsoidal and proposals are uniform within the constrained prior. The SH0ES and Planck seeds are 2301 and 2401, and the stopping threshold is $\Delta \ln Z = 0.01$ [13]. The likelihood calls the exact Palatini roots and full covariance terms. Equal-weight posterior samples are obtained by importance resampling; reported intervals are the 16th, 50th, and 84th percentiles. The evidence values from the two runs are not a Bayes factor because the runs condition on different data likelihoods.

5. Results and anchor consistency

Table 1 reports the independent-anchor posterior constraints.

5.1. Posterior elongation

The two posterior sample sets give

$$\rho_{nk}(\text{SH0ES}) = 0.8406, \quad (31)$$

$$\rho_{nk}(\text{Planck}) = 0.8474, \quad (32)$$

$$(a_{\text{major}}/a_{\text{minor}})_{\text{SH0ES}} = 9.19, \quad (33)$$

$$(a_{\text{major}}/a_{\text{minor}})_{\text{Planck}} = 8.98, \quad (34)$$

where the axis ratios are derived from the eigenvalues of the posterior sample covariance in native (n, k) coordinates. The major-direction slopes are $\Delta k/\Delta n = 0.1762$ and 0.1861 . Thus increasing n by 0.1 can be compensated locally by increasing k by approximately 0.018.

The origin is both physical and statistical. Increasing n changes the redshift evolution of the inverse-curvature term in Eq. (7), while increasing k strengthens the dust source. Re-computing $\bar{\beta}$ to maintain $E(0) = 1$ lets these effects produce similar low-redshift expansion histories. Supernovae constrain an integral of $1/E(z)$ and marginalization over q removes their absolute scale. The chronometers and one H_0 anchor provide direct expansion-scale information, while DESI adds transverse and radial distance-shape information through $z = 2.33$. DESI therefore contracts the former long ridge into a much smaller tilted ellipse, although the remaining correlation shows that the same physical compensation is still active.

At the maximum-marginalized-likelihood samples, the BAO contributions are $\chi_{\text{BAO}}^2 = 10.110$ and 10.515 for SH0ES and Planck, respectively. These are conditional component residuals at jointly selected (n, k) values, not independently calibrated goodness-of-fit statistics. A separate DESI-only minimization over (n, k, A_{BAO}) gives $(n, k, A_{\text{BAO}}) = (-0.057, 0.291, 29.690)$ and $\chi^2/\nu = 9.855/10$, for a probability to exceed of 0.453. Thus the parameter-counted BAO-only fit shows no shape inconsistency. The joint posterior scales are $A_{\text{BAO}} = 29.849^{+0.235}_{-0.237}$ and $29.955^{+0.246}_{-0.237}$. Combining their conditional draws with the alternative H_0 anchors gives $r_d = 138.653^{+2.080}_{-1.978}$ Mpc and $148.303^{+1.581}_{-1.552}$ Mpc. Their difference is largely the algebraic mapping of the discrepant anchors through $r_d = c/(H_0 A_{\text{BAO}})$; it is not a self-consistent early-Universe determination or an independent tension test.

For SH0ES and Planck, respectively, the posterior probabilities of $n > 0$ are 0.070 and 0.042. Both central 68 percent intervals lie below $n = 0$, but the corresponding $P(n < 0) = 0.930$ and 0.958 represent only modest one-sided background-level preferences, not compelling evidence for a departure from Λ CDM. The derived $\bar{\beta}$ distribution remains mildly skewed because Eq. (14) maps the elongated (n, k) posterior nonlinearly.

To isolate the DESI update, we compare with matched no-DESI runs having the same priors, chronometer and Pantheon+ data, anchors, sampler settings, and integer seeds. For SH0ES, DESI reduces the central 68 percent widths of n and k by factors of 4.63 and 7.18 and moves their medians by 0.81 and 0.76 of the old posterior standard deviations. For Planck, the corresponding contraction factors are 4.96 and 7.01, while the median shifts are only 0.04 and 0.05 old standard deviations. The H_0 widths and medians are virtually unchanged because the external anchors dominate that scale. Since each with/without pair

Table 1: Posterior medians and 16th–84th percentile intervals from the Pantheon+ + 32-chronometer + DESI DR2 dynesty runs. The reported r_d is derived from $c/(H_0 A_{\text{BAO}})$ and is conditional on the nuisance-scale prior, not an early-Universe Palatini prediction. The final column is a conventional conditional-mean χ^2/ν diagnostic evaluated at the maximum-marginalized-likelihood sample; it is not used to define the credible intervals.

Anchor	β	n	$k = \Omega_{m0}$	H_0 [km s ⁻¹ Mpc ⁻¹]	M_B	A_{BAO}	r_d [Mpc]	χ^2/ν
SH0ES	3.495 ^{+0.450} _{-0.387}	-0.098 ^{+0.067} _{-0.067}	0.287 ^{+0.014} _{-0.013}	72.444 ^{+0.947} _{-0.964}	-19.281 ^{+0.028} _{-0.029}	29.849 ^{+0.235} _{-0.237}	138.653 ^{+2.080} _{-1.978}	1430.118/1631
Planck base- Λ CDM	3.373 ^{+0.431} _{-0.380}	-0.116 ^{+0.067} _{-0.067}	0.287 ^{+0.014} _{-0.015}	67.471 ^{+0.490} _{-0.481}	-19.433 ^{+0.017} _{-0.016}	29.955 ^{+0.246} _{-0.237}	148.303 ^{+1.581} _{-1.552}	1428.430/1631

shares all non-BAO observations, these are descriptive update measures, not independent-data tension statistics. Absolute evidence values across likelihoods of different dimension are likewise not used as a DESI inclusion statistic.

5.2. Independent-anchor consistency with IOI and Q_{DM}

For two independent Gaussian constraints with mean difference $\delta = \mu_1 - \mu_2$ and broad common priors, the difference-in-means statistic is [15]

$$Q_{\text{DM}} = \delta^T (C_1 + C_2)^{-1} \delta, \quad (35)$$

where C_1 and C_2 are the covariance matrices associated with the mean estimates μ_1 and μ_2 . Independence makes the covariance of their difference equal to $\text{Cov}(\delta) = C_1 + C_2$. For the present scalar H_0 comparison, $C_1 = \sigma_{\text{SH0ES}}^2 = (1.04)^2$ and $C_2 = \sigma_{\text{Planck}}^2 = (0.50)^2$, in units of (km s⁻¹ Mpc⁻¹)². For correlated estimates the difference covariance would instead be $C_1 + C_2 - C_{12} - C_{21}$, where C_{12} and C_{21} are cross-covariance blocks; Eq. (35) in its displayed form therefore applies only when those blocks vanish. Under the Gaussian assumptions, Q_{DM} follows a chi-square distribution with degrees of freedom equal to the rank of $C_1 + C_2$. The two-experiment index of inconsistency is [14]

$$\text{IOI} = \frac{1}{2} Q_{\text{DM}}. \quad (36)$$

For the one-dimensional published H_0 summaries,

$$Q_{\text{DM}} = \frac{(73.04 - 67.4)^2}{1.04^2 + 0.5^2} = 23.888, \quad (37)$$

$$\text{IOI} = 11.944, \quad (38)$$

with one degree of freedom, $p = 1.021 \times 10^{-6}$, and an equivalent two-sided Gaussian separation of 4.888σ . In this one-dimensional Gaussian limit, IOI and Q_{DM} are algebraically related summaries, not independent pieces of evidence.

Crucially, the value 23.888 is fixed entirely by the two published input anchors (73.04 ± 1.04) and (67.4 ± 0.5) before either Palatini likelihood is evaluated. It would be identical for any downstream gravity model using those same Gaussian summaries. It therefore quantifies the input-anchor tension; it is not a tension generated, amplified, or resolved by the Palatini model.

Applying Eq. (35) naively to the two full Palatini posterior covariances would be incorrect because those posteriors share Pantheon+ and chronometer and DESI data; their parameter differences have a nonzero cross-covariance. We therefore evaluate the standard statistics on the genuinely independent anchor inputs. The Planck number assumes base- Λ CDM rather than a self-consistent Palatini CMB calculation, so its interpretation remains conditional on that background model.

6. Verification and limitations

We reconstructed the physical branch for every one of the 24739 retained raw nested points (12394 SH0ES and 12345 Planck) on 234 redshifts spaced by $\Delta z = 0.01$ over $0 \leq z \leq 2.33$, which covers the Pantheon+ interval $0 \leq z \leq 2.26$. There were zero failures. The smallest g' , F , and ξ values were respectively (0.01599, 0.01638, 0.75381) for SH0ES and (0.03214, 0.05916, 0.75280) for Planck; the largest relative trace residual was 1.01×10^{-12} and the largest $|E^2(0) - 1|$ was 1.28×10^{-12} . These retained points include the low-weight prior-edge trajectory, not only posterior-resampled draws. The proof following Eq. (7) applies throughout the full open prior, and numerical evaluations that fail a physical-domain condition are excluded rather than reassigned to a different algebraic branch.

Independent numerical checks cover the exact $n = 0$ limit, $E(0) = 1$ on multiple branches, regular evaluations arbitrarily close to the prior endpoints, luminosity and BAO distances against adaptive high-accuracy quadrature, the observational selections and covariance matrices, synthetic likelihood and acoustic-scale recovery, analytical nuisance conditionals, and the exact one-dimensional IOI/ Q_{DM} relation. At the two DESI posterior medians, increasing the distance grid from 700 to 1400 points changes the marginalized log likelihood by only 0.0031 (SH0ES) and 0.0026 (Planck); increasing it again to 2800 changes the result by only 0.00022 and 0.00011. The production evidence-error estimates are 0.076 and 0.075 in $\ln Z$; these values are sampler diagnostics and are not used to compare the alternative anchors.

The with- and without-DESI comparisons use the same non-BAO observations, priors, and numerical settings. The BAO-only goodness-of-fit calculation uses 13 measurements minus three fitted parameters (n, k, A_{BAO}). It is a focused shape diagnostic, whereas the displayed global χ^2/ν is only conventional because correlated Pantheon+ light curves are not independent scalar observations.

Several limitations remain. Chronometer systematics continue to depend on stellar-population modeling. New LEGA-C analyses cannot simply be added when they reuse galaxies already represented in the 32-point compilation. Pantheon+ light curves are correlated and can include repeat observations, so a naive data-count degree of freedom is only a conventional goodness-of-fit summary. DESI constrains only late-time distance ratios here: the free A_{BAO} scale avoids an inconsistent Λ CDM calibration but does not replace a Palatini calculation of pre-recombination physics or r_d . The analysis omits a self-consistent Palatini CMB likelihood, radiation, growth, and perturbation or local-gravity viability tests. Finally, the Planck-

conditioned run is a sensitivity calculation, not a complete early-Universe constraint on this gravity theory.

7. Conclusion

We have updated the Palatini calculation to use Pantheon+, a 32-point cosmic-chronometer likelihood with available systematic covariance, and the 13-element DESI DR2 BAO likelihood. DESI is included as an uncalibrated late-time ruler by marginalizing $A_{\text{BAO}} = c/(H_0 r_d)$ rather than importing a Λ CDM sound horizon. The observable matter parameter is $k = \kappa \rho_{m0}/(3H_0^2)$, while $E(0) = 1$ derives $\bar{\beta}$ from (n, k) . Analytical marginalization over the conditional H_0 , supernova-intercept, and BAO-scale directions makes full-covariance nested sampling practical and yields bounded, asymmetric posterior intervals.

DESI contracts the positive-slope (n, k) degeneracy substantially: relative to matched no-DESI runs, the central widths of n shrink by factors of 4.6–5.0 and those of k by about 7. Both central intervals lie below $n = 0$, but the one-sided probabilities of 93.0 and 95.8 percent are not strong background-level evidence against Λ CDM. The acceptable BAO-only probability, 0.453, shows that the tightening is not accompanied by an obvious distance-shape misfit. The conditional r_d values must not be interpreted as early-Universe Palatini predictions. The H_0 anchors themselves have $Q_{\text{DM}} = 23.888$, $\text{IOI} = 11.944$, and $p \approx 10^{-6}$. Reporting those standard inconsistency statistics quantifies the discrepancy between the published anchor summaries. The next scientific step is a matched Palatini-versus- Λ CDM analysis with a theory-consistent sound horizon and CMB likelihood, alternative prior measures, and perturbation viability.

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Data and code availability

The observational inputs and covariance information are available from the public releases cited above. The analysis code and derived posterior products needed to reproduce the reported results accompany this manuscript.

AI-tool disclosure

AI-based writing and analysis tools were used during the preparation of this manuscript. All generated text and contextual information were independently checked and refined by the authors.

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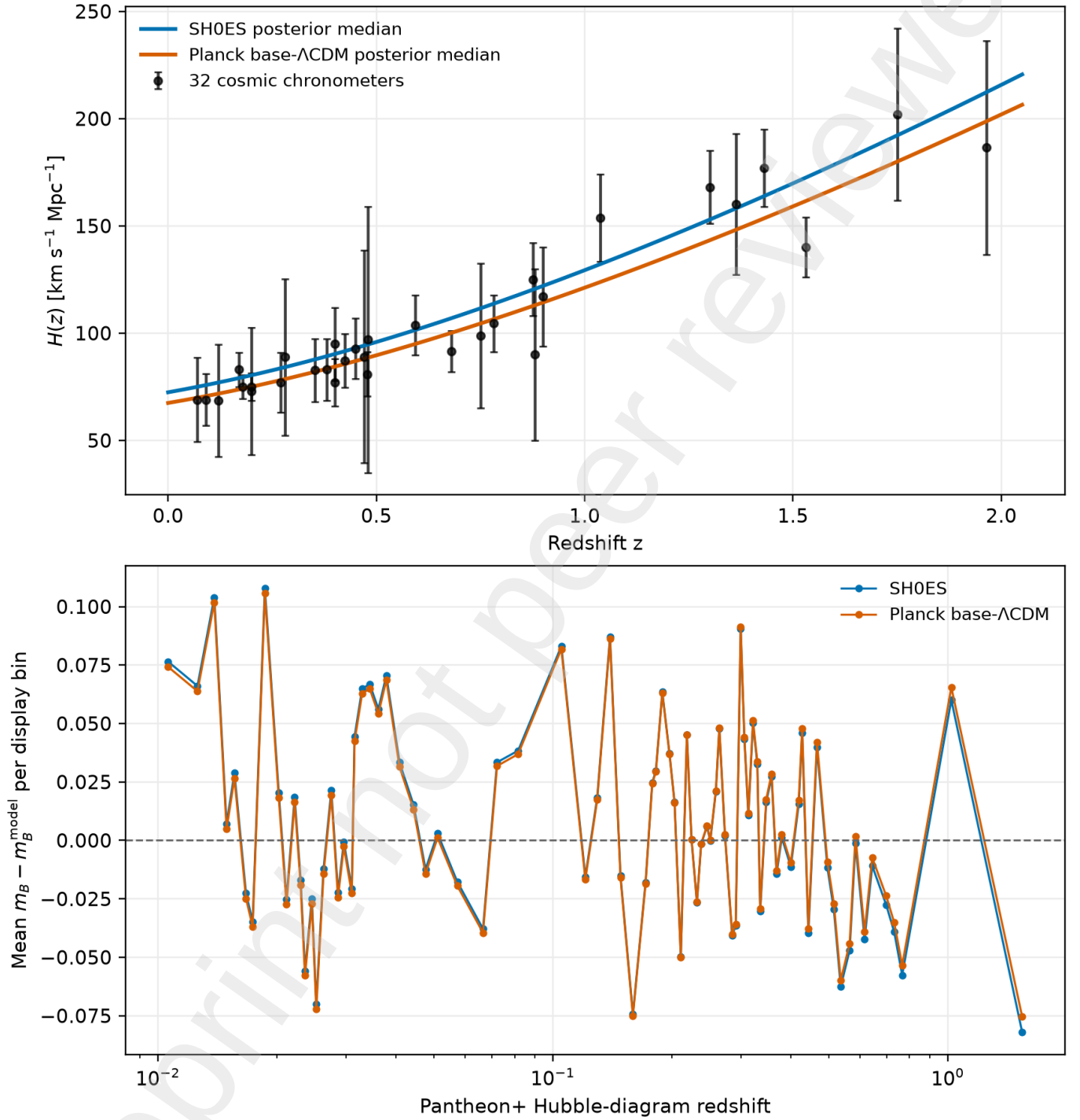


Figure 1: Top: the 32 cosmic-chronometer measurements, with diagonal error bars shown for visualization, and posterior-median Palatini expansion histories for the two independent anchors. The likelihood uses the full chronometer covariance. Bottom: display-binned Pantheon+ magnitude residuals; all 1590 selected light curves and the complete released covariance enter the fit.

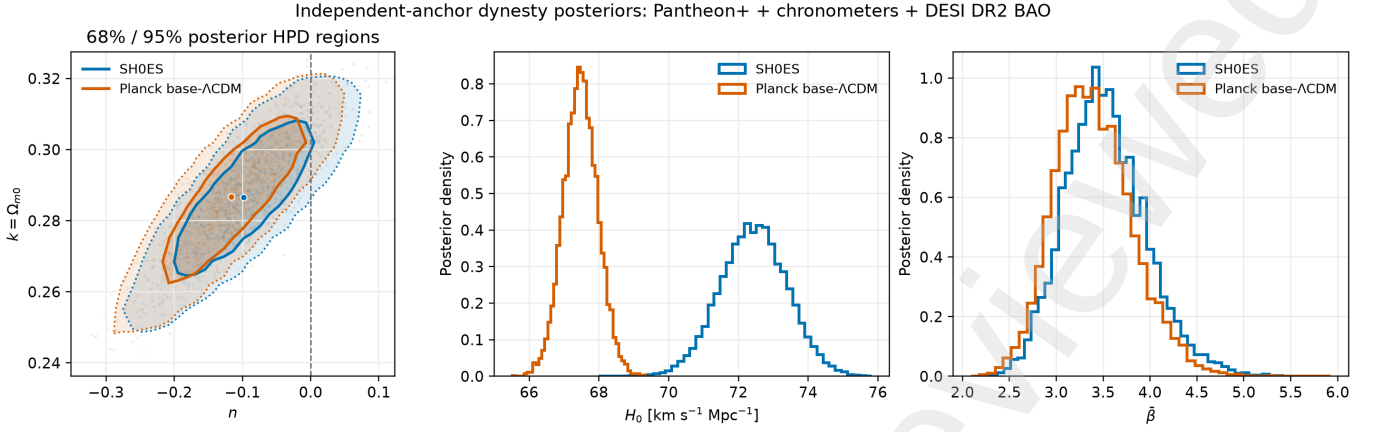


Figure 2: Independent-anchor posterior comparison. Left: 68 and 95 percent highest-posterior-density regions in (n, k) from a Gaussian kernel density estimate of the posterior-weighted nested samples. Reflection corrects density loss at the lower- n prior boundary; faint points show weighted posterior support, circles mark medians, and the dashed line marks the exact $n = 0$ Λ CDM background limit. The display window contains both 95 percent HPD supports, while the KDE and HPD thresholds are computed on the full sampling-prior domain. Center and right: marginalized H_0 and derived β distributions. DESI sharply contracts the (n, k) region compared with the no-DESI analysis while retaining a visible positive-slope elongation.

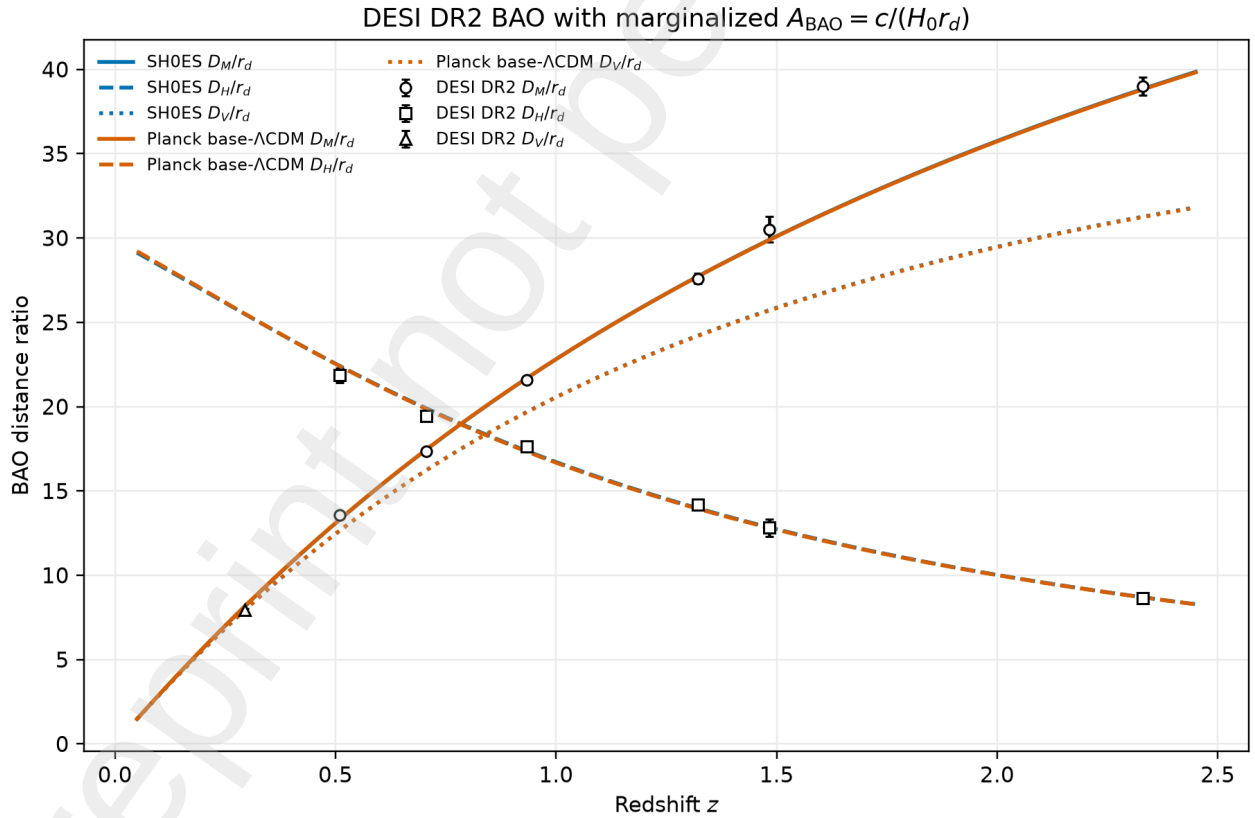


Figure 3: The 13 DESI DR2 BAO ratios and posterior-median Palatini distance predictions. Error bars show the covariance diagonal for visualization; the likelihood uses the complete released covariance. Line style identifies D_M/r_d , D_H/r_d , or D_V/r_d , and color identifies the late-time H_0 anchor. The common scale A_{BAO} is marginalized independently in each run, so these curves do not assume a Λ CDM sound horizon.